

Detection of Mayer waves in non-regularly sampled data using Lomb-Scargle periodograms

1

Victor Hugo de Souza S. Ragazzi¹, Leonardo Bonato Felix¹, Hasana Bagnall Hare² and John Vincent Ringwood²

¹ Federal University of Viçosa/Department of Electrical Engineering, FAPEMIG, Viçosa, Brazil

² Maynooth University/Department of Electronic Engineering, Maynooth, Ireland

Abstract— The present work proposes the use of the Empirical Mode Decomposition (EMD) method conjointly with the Lomb-Scargle (LS) periodogram for the identification of Mayer waves (low-frequency oscillations in blood pressure). The EMD-LS method aims to overcome the limitations of the Fast Fourier Transform (which requires the signal to be periodically sampled), and to find the best pre-processing configuration to detect this waves in noisy signals and with data loss. The work also presents the obtained results with different combinations of Intrinsic Mode Functions (IMF) and shows that the combination of the average of IMF 1 and 2 achieved the highest detection accuracy (87.5%) on the calibration database. In addition, it demonstrates that the EMD-LS method was effective and did not show a significant difference in relation to the test using the complete database (even with the removal of 35% of the original data). It also highlights the importance of detecting those waves, which can provide relevant information about the patient's condition and be used in testing models of the neural baroreflex system.

Keywords— Mayer waves; Lomb-Scargle periodogram; Empirical Mode Decomposition; Low frequency; Blood pressure.

I. INTRODUCTION

There is a specific interest in the identification of the so-called ‘Mayer waves’, which are low-frequency oscillations in blood pressure, during investigations into blood pressure variations. These waves stem from oscillations in the control systems of baroreceptor and chemoreceptor reflexes [1]. These reflexes arise from the activity of the sympathetic autonomic nervous system, which plays a role in regulating blood pressure. These waves manifest themselves in various frequency bands, varying between different species. In humans, they occur in the 0.1 Hz range [2, 3].

Patients with ventricular assist devices (VAD) are ideal for analysis as they present more Mayer waves, however, the reduction in pulsatility caused by the device increases the difficulty of obtaining precise and non-invasive BP records.

Most non-invasive BP measurement devices (including the one used in the study) record the BP at the signal's peak, which means that the data are irregularly sampled. In addition, the reduced pulsatility observed in patients with VAD leads to a reduced BP pulse height [1], which means that the device may not be able to identify some peaks, leading to missing data points. Initially, we might assume only a few data points are lost. However, this study reveals that approximately 60-80% of data is missing for some patients.

To identify Mayer waves, the Fast Fourier Transform (FFT), widely used in various areas due to its robustness and ease of implementation, could be a viable option [4]. On the other hand, the FFT assumes that the analyzed signal repeats over time, which is not always the case with biological signals like blood pressure, which leads us to seek other alternatives. Studies indicate that signal interpolation can be a viable solution to this problem [5]. However, it is necessary to perform the interpolation correctly, as this method can introduce noise into the signal spectrogram, especially in the low frequency range, where Mayer waves appear [6].

On the scientific literature, we find studies that indicate that periodograms, such as, for example, the Lomb-Scargle (LS) methodology or the Welch method, may be more effective than the FFT for noisy signals from tachograms [5, 7]. The LS methodology, after all, assigns a weight to each data point based on its time variance; this approach doesn't require resampling the data, making it suitable for analyzing non-periodic or unevenly sampled signals. Originally developed for astrophysical applications, the LS periodogram is an established procedure for generating a power spectrum. This spectrum is used to detect and characterize the periodic components of a signal [8].

Given the need to detect such a phenomenon in BP signals with irregular sampling, this study employs the LS methodology combined with the pre-processing already presented in the literature and used in this context [7, 9]. This pre-processing is known as Empirical Mode Decomposition (EMD), a method that allows the decomposition of a time se-

ries into oscillatory components, called Intrinsic Mode Functions (IMF) [10, 11].

The significance of the Mayer wave phenomenon in the human body is an ongoing area of research. While the origins of these waves are not fully understood, studies suggest they play a significant role in our biological system [2, 3]. In this context, the aim of this study is to contribute to the detection of these waves, allowing the validation of models related to the neural baroreflex system [1]. Future investigations are expected to better reveal the importance of this phenomenon in our body. The main challenge for the analysis to be carried out is the identification of low frequencies in the PA signal, even in the face of a massive data loss that can occur.

II. MATERIALS AND METHODS

A. Data

The blood pressure data were obtained from the EuroBaVar Dataset. In this study, volunteers participated in two recording sessions conducted in a quiet room, with a constant temperature of 20 °C and controlled lighting. During each session, the participants were monitored through non-invasive sensors and electrodes, which continuously recorded blood pressure. In addition, participants were asked to avoid speaking and moving during the recordings, and they did not have access to the examination equipment screen.

The first recording session was conducted with the participant in an upright position, after a 5-minute adaptation period. The second session was conducted with the participant in a supine position, after a 10-minute rest period.

The collected data were stored in ASCII text files, tab-separated, resulting in a total of 21 files for each type of position. Each file was identified by an alphanumeric code, which indicated the participant's number and position (L for upright position and S for supine position).

In this database, it was observed that patients 4, 11, 14, 16, and 17 manifested the Mayer waves while in an orthostatic position, while patients 5, 6, 7, 10, and 12 did not present these waves during the supine position. These data were then used in the calibration of the model.

B. Detection Strategies

In the subsequent sections, a more detailed description of the two proposed methodologies for the detection of Mayer waves is carried out. Initially, we will explore the gold standard method, involving Fourier analysis paired with interpolation techniques [12]. The second solution is the EMD-LS

method. Each of these approaches has its peculiarities and effectiveness, which will be discussed and compared in detail in the upcoming sections of this article.

B1 FFT Analysis Model

In the process of identifying Mayer waves, a conservative approach was employed to ensure that the resulting analysis was free from incorrect identifications. This approach consists of an FFT analysis conducted by the researchers. Interpolation was used as a tool to fill any gaps in the data. Additionally, the analysis was complemented with the application of an FFT. This technique allowed for the analysis of the signal's spectrogram.

For interpolation, the numerical method known as Spline was used. This method aims to approximate functions by dividing the interpolation interval into several subintervals. In each of these subintervals, a low-degree polynomial is adjusted. In this study, the cubic Spline form was chosen, which employs third-degree polynomials. The choice of this form of interpolation, over linear interpolation, results in a more pronounced smoothing of the data. Please note that this translation maintains the terms and didactics of the original text, with changes made only where necessary for better comprehension.

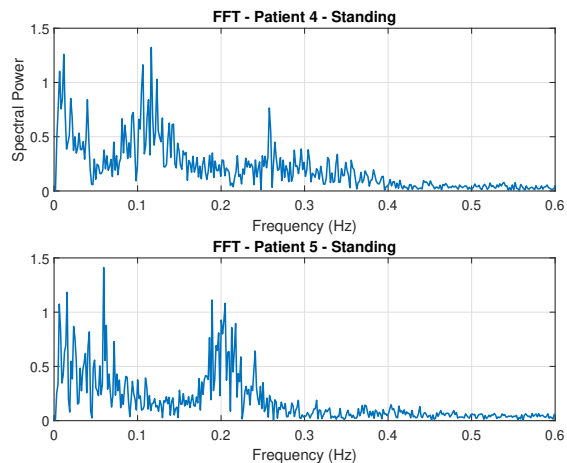


Fig. 1: Application of the FFT to data with interpolation. Patient 4 presents Mayer waves while patient 5 does not.

The step adopted in the interpolation was 0.66 seconds. After normalizing the data, an FFT is applied, resulting in the spectrogram of the data. The Figure 1 illustrates this spectrogram, specifically of Patient 4 in an orthostatic position and Patient 5 in a supine position. The responses obtained through this analysis will be used as a template for

comparing the responses obtained by the proposed model below.

B2 EMD-LS model

a) *Preprocessing*: The EMD method is a technique that allows decomposing a non-stationary and non-linear signal into a set of intrinsic mode functions (IMF) [7, 10, 11]. These are simple oscillations with variable amplitude and frequency. An IMF is defined as a function that satisfies two conditions:

- The number of extremes (maximums and minimums) and the number of zero crossings must be equal or differ by at most one;
- The mean of the upper and lower envelopes, defined by the local extremes, must be zero at all points.

The residue corresponds to the part of the signal that cannot be decomposed into any IMF, that is, it represents a trend or a constant. The empirical mode decomposition method can extract the low-frequency components of blood pressure signals by applying an iterative algorithm that consists of the following steps:

- Identify all local extremes of the original signal;
- Interpolate between the local maxima to obtain the upper envelope, and between the local minima to obtain the lower envelope;
- Calculate the mean of the upper and lower envelopes, and subtract from the original signal, obtaining a possible IMF candidate;
- Check if the IMF candidate satisfies the defined conditions, otherwise, repeat the previous steps using the candidate as the input signal;
- If the IMF candidate satisfies the conditions, extract this component from the original signal, obtaining a residue;
- Repeat the process with the residue until it becomes a trend or a constant, or until the number of IMFs extracted reaches a predefined limit;
- The result is a decomposition of the original signal into a set of IMFs and a residue, which can be added together to reconstruct the original signal. The lower order IMFs correspond to the high frequency components of the signal, while the higher order IMFs correspond to the low frequency components of the signal.

In this study, all variations of IMF were tested, using IMF 1, 2 and 3 alone and also the averages between them, to find the combination that presents the pre-processing with the highest accuracy when applying the calibration database.

b) *Lomb-Scargle Periodogram*: The Lomb-Scargle periodogram is an algorithm for detecting and characterizing periodic signals in irregularly sampled data. It is based on the idea of fitting a sinusoidal function to the data and calculating the power of the fit for each oscillation frequency. Power is a measure of how well the sinusoidal function fits the data, and the greater the power, the greater the evidence of periodicity [8].

The method is a generalization of the FFT power spectrum, which is only applicable to uniformly spaced data. The Lomb-Scargle periodogram considers the observation times of the data and allows estimating the power for arbitrary frequencies, without the need to interpolate or fill in the data.

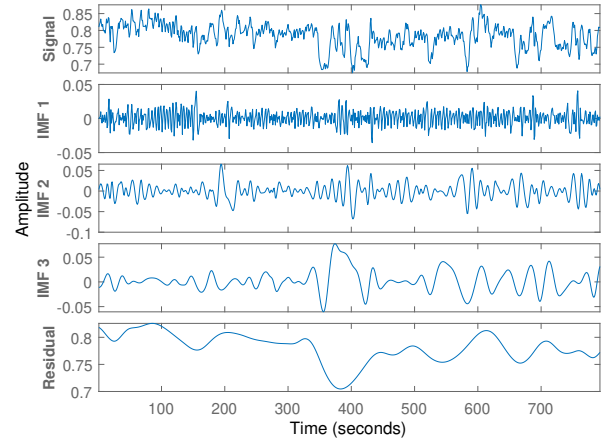


Fig. 2: The decomposition in empirical mode is presented up to the third decomposition to reconstruct the patient's original signal 5.

After empirical mode decomposition, the PA signals were subjected to the Lomb-Scargle periodogram, using a maximum frequency of 0.4 Hz. The powers obtained were normalized by twice the variance of the data, in order to avoid discrepancies arising from the different amplitudes of Mayer waves among individuals. A critical value was calculated that defines the spectral power threshold, so that a peak above this value has a 1% probability of being a true signal peak and not the result of random fluctuations. To consider that there was detection of Mayer waves, the band of 0.08–0.13 Hz was used, which is close to the frequency of 0.1 Hz, this being the frequency where the phenomenon appears in humans.

Finally, after the application of the methods described, points will be randomly removed from the data and, in this way, discover how far the tools used in this study can detect Mayer waves.

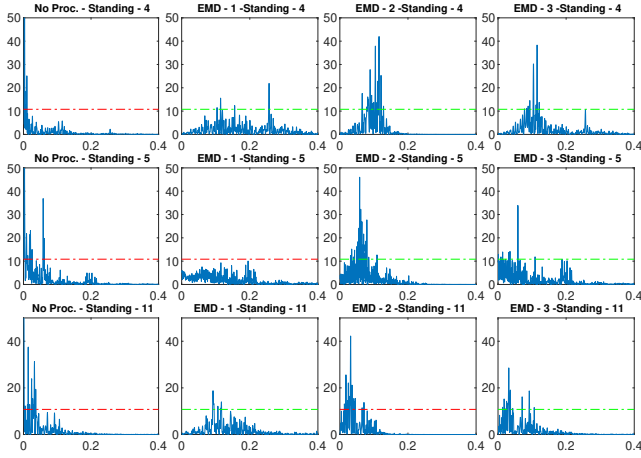


Fig. 3: The periodogram was applied to a group of 10 patients, only 3 of these are represented in the figure. The standing patients have the presence of Mayer waves while those in the supine position do not. The dotted line represents the critical value of the periodogram, the red lines are the tests that did not predict the presence of Mayer waves and the green lines are the present predictions. The number at the end of each graph corresponds to the patient number.

III. RESULTS

The Figure 2 presents the IMF curves decomposed up to 3 decompositions. To define which decompositions would be used for the complete analysis of the data, a part of the database was used for testing to adjust a possible better combination between the decomposed curves, using the curves alone or the average between them.

Applying each possible configuration for signal reconstruction as preprocessing, the database was used for calibration in order to analyze which configuration results in better accuracy. Applying the Lomb-Scargle periodogram, the results presented in the Figure 3 were obtained. Based on previous analyses, it was found that composition 3, obtained by the average of IMFs 1 and 2, achieved a detection accuracy of 90% in the calibration database.

Continuing with the analysis, composition 3 was applied to the entire data set. Subsequently, the periodogram was used on the processed data set, resulting in the generation of the Table 1. This table indicates the presence or absence of Mayer waves in each patient, both for FFT analysis evaluation and for the analysis of the proposed model. Through the comparison between both, it is possible to evaluate the effectiveness of the model in correctly identifying the phenomenon.

The Table 2 demonstrates the efficacy of the tested method by reducing the amount of data used in the tests. Using the

McNemar test, it was possible to calculate that the results began to show a significant difference after the removal of 35% of the original database. In other words, the complete data achieved the same efficacy as only 65% of the randomly selected data, with a significance of 95%.

IV. DISCUSSIONS

Table 1 shows that positive detections occurred in 12 patients in the orthostatic position and 10 detections in patients in the supine position. It was observed that some patients presented peaks in the 0.1 Hz range, however, the origin of these peaks was difficult to determine, as it was not clear whether they were caused by the phenomenon under study or by noise. This statement can be observed in the Figure 1.

In the spectrogram resulting from the Fourier analysis, presented in the Figure 1, there is an uncertainty about the presence of Mayer waves that cannot be assured in patient 5. This uncertainty is probably attributed to the noise introduced by the interpolation method used in the frequency range, which ends up hiding the peak associated with this

Table 1: Identification of the presence or absence of Mayer waves in the database. The patients who present “*” are those used for the model calibration.

Patient	EMD-LS Model		FFT analysis	
	Standing	Lying	Standing	Lying
1	Yes	Yes	Yes	Yes
2	Yes	Yes	Yes	Yes
3	Yes	No	Yes	No
4*	Yes	Yes	Yes	Yes
5*	Yes	Yes	No	No
6*	No	No	No	No
7*	Yes	No	No	No
8	Yes	No	Yes	No
9	Yes	No	Yes	No
10*	No	No	No	No
11*	Yes	No	Yes	Yes
12*	No	No	No	No
13	No	Yes	No	No
14*	Yes	Yes	Yes	Yes
15	No	No	No	Yes
16*	Yes	No	Yes	No
17*	Yes	Yes	Yes	Yes
18	No	No	No	Yes
19	Yes	Yes	Yes	Yes
20	No	No	No	No
21	Yes	No	Yes	Yes

Table 2: Data demonstrating the efficiency of the applied model corresponding to data loss. The table presents the accuracy values, the value of the statistical test used to verify the significance of the model applied to missing data, and the F1-Score value. The data marked with “*” are those with significant differences.

Data loss (%)	Number of detections: Standing	Number of detections: Lying	Accuracy (%)	McNemar Value	F1-Score
Ground Truth	12	10	100	1	1
0	14	8	81.8	0.25	0.818
5	14	8	81.8	0.25	0.818
10	14	9	81.8	0.25	0.800
15	14	8	81.8	0.25	0.818
20	11	8	77.3	0.5	0.829
25	10	8	72.7	0.25	0.800
30	12	7	72.7	0.25	0.781
35	12	7	68.2	0.0625	0.732
40*	9	4	45.5	0.03125	0.571
45	8	5	50	0.0625	0.629
50*	8	4	31.8	>0.01	0.412
55*	5	3	27.3	>0.01	0.400
60*	4	5	27.3	>0.01	0.387
65*	5	2	27.3	>0.01	0.414
70*	2	1	9.1	>0.01	0.160

phenomenon. However, this peak can be detected when examining the spectrogram generated by the Lomb-Scargle periodogram of the same patient, seen in the Figure 3.

In the empirical mode decomposition, it is observed that the first IMF corresponds to the highest frequencies, which is reflected in the second column of Figure 3, where the powers of the LS periodogram rarely cross the established threshold. The second IMF has a larger amplitude in the analyzed frequency range, presenting high peaks in the periodogram powers, which generate false detections in the data that do not contain true responses for this frequency.

The solution of using the average between the two IMFs emerged as a balance between the two compositions, reducing peaks of false responses and highlighting the peaks of real responses, resulting in an accuracy of 90% of the test in the calibration database.

For the calculation of the LS periodogram threshold, a false-positive rate of 1% was used. This value may justify the low detection of Mayer waves in patients in the supine position, as a high threshold reduces the probability of false positives, but decreases the sensitivity of detection.

Furthermore, it should be considered that the blood pressure of patients in an orthostatic position may increase due to gravity, which can amplify the amplitude of low-frequency waves, making them more visible in this position.

The methods were applied to the data with random point loss, with a progression of 5 to 5% loss until the Mayer waves could no longer be determined in the data. This analysis is

shown in the Table. 2.

For the incomplete data, the McNemar test was performed using the complete data as a reference. The results showed significant differences only after a loss of 35% of the total data, where it was observed that the method went from 22 to 19 detections, resulting in 13% less accuracy. Furthermore, it is important to note that the F1-score remains above 0.7, a relatively high value, especially for applications that are still significantly similar. These findings indicate that the method continues to be effective even when there is a considerable loss of data.

However, the Mayer waves could still be predicted using only 30% of the complete data. For a smaller amount of data, this was no longer feasible.

V. CONCLUSIONS

With the detection of Mayer waves (low-frequency waves) in blood pressure monitoring, it is possible to identify the use of AVDs, or even pathologies in the individual such as lack of pulsatility. Even though the consequences of Mayer waves are not well-defined in current literature, their causes are defined [2, 3] and can provide information about important patient conditions [13, 14] or even verifications of neural baroreflex system models [1].

Even with this range of possibilities for potential diagnostic and prognostic revelations, heart rate analyses do not have significant practical applications. This may be due to the time

spent tracing tachograms and cardiograms, or even the difficulty in understanding the results, which require an interdisciplinary load, going beyond the usual clinical scope [7].

It is noteworthy that, compared to the FFT analysis methodology employed, the EMD-LS model demonstrated remarkable performance. This model presented high accuracy and F1-Score indices, even after the removal of a large part of the data set. In addition, the spectrogram generated by the EMD-LS model provides a clearer visualization, as it does not rely on interpolation methods, which are responsible for adding noise in the frequency bands of interest in this study. Therefore, the absence of such interpolation methods in the EMD-LS model contributes to the clarity and precision of the results obtained.

However, the frequency range in which Mayer waves manifest in each being varies according to specific physiological characteristics, including the size of the individual. It is notable that in rodents, for example, these waves have a frequency of 0.4Hz. In beings of the same species, a significant variation can be observed. In this study, a large frequency range was addressed, in an attempt to generalize the occurrence for beings of all sizes. However, it is important that future studies make a more precise adjustment in the frequency range for each individual.

With the proposed tools, this study aims to enable the application of these methods in the clinical context for deeper analyses about the possible diagnoses that can be interpreted from data obtained in blood pressure screenings.

CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

ACKNOWLEDGEMENTS

The described work was funded by the FAPEMIG institution in the scientific initiation program by UFV. The authors particularly thank the members of the NIAS research group, Alexandre Gomes Caldeira and Pedro Lucas de Oliveira Mateus for the discussions on the topic and corrections pointed out in the writing of the text.

REFERENCES

1. Ringwood John V.. Understanding the interplay between baroreflex gain, low frequency oscillations, and pulsatility in the neural baroreflex *Biocybernetics and biomedical engineering*. 2020.
2. Julien Claude. The enigma of Mayer waves: Facts and models *Cardiovascular Research*. 2006.
3. Julien Claude. An update on the enigma of Mayer waves *European Society of Cardiology*. 2019.
4. Li Kai. Spectral Analysis of Heart Rate Variability: Time Window Matters *Frontiers in Neurology*. 2019.
5. Tihak Amina. Explorative Evaluation of Heart Rate Variability Spectral Analysis Approaches *IEEE*. 2022.
6. Morelli Davide. Analysis of the Impact of Interpolation Methods of Missing RR-Intervals Caused by Motion Artifacts on HRV Features Estimations *Sensors*. 2019.
7. Stewart Jill. Application of the Lomb-Scargle Periodogram to Investigate Heart Rate Variability during Haemodialysis *Journal of Healthcare Engineering*. 2020.
8. Plas Jacob T Vander. Understanding the Lomb-Scargle Periodogram *The Astrophysical Journal Supplement Series*. 2018.
9. Han G.. Electrocardiogram signal denoising based on empirical mode decomposition technique: an overview *IOP Publishing Ltd and Sissa Medialab*. 2017.
10. Souza Neto Edmundo Pereira. Empirical Mode Decomposition to assess cardiovascular autonomic control in rats *Fundamental Clinical Pharmacology*. 2007.
11. Mert Ahmet. Detrended fluctuation thresholding for empirical mode decomposition based denoising *Digital Signal Processing*. 2014.
12. Li Kai. Spectral Analysis of Heart Rate Variability: Time Window Matters *Front. Neurol.*. 2019.
13. Fukuta Hidekatsu. Prognostic value of heart rate variability in patients with end-stage renal disease on chronic haemodialysis *Nephrol Dial Transplant*. 2003.
14. MG Barnas. Hemodynamic patterns and spectral analysis of heart rate variability during dialysis hypotension *J Am Soc Nephrol*. 1999.

Author: Victor Hugo de Souza Singulani Ragazzi
Institute: Federal University of Viçosa
Street: Travessa José Valentino Cruz
City: Viçosa
Country: Brazil
Email: victorhugosouza0@gmail.com